Notes: Kosowski, Timmermann, Wermers, and White (JF, 2006)

Can Mutual Fund “Stars” Really Pick Stocks? New Evidence from a Bootstrap Analysis

Contents

1 Big picture

- **Research question.** Are extreme mutual fund winners genuine stock-pickers, or are they simply the lucky winners in a very large cross section of funds? More specifically, how many high-alpha funds should exist by chance once the econometrician accounts for nonnormal fund returns, heterogeneous risk-taking, and the ex post ranking of many funds?
- **Main idea.** Looking at the t-statistic of a single top fund is misleading because the fund was selected after sorting hundreds or thousands of funds. The correct benchmark is the distribution of the best, second-best, and percentile-ranked fund under a no-skill world.
- **Core method.** The paper estimates fund alphas, resamples fund residuals, imposes the null of zero true alpha, reconstructs artificial fund returns, and repeatedly sorts the artificial cross section. This bootstrap produces the luck benchmark for each ranked position in the alpha distribution.
- **Main finding.** A sizable minority of mutual fund managers, especially growth-oriented managers, appear to pick stocks well enough to cover expenses and trading costs. At the same time, many poor-performing funds significantly underperform net of costs.
- **Key contrast with earlier work.** Earlier studies often concluded that the average mutual fund underperforms and that apparent stars are not very persistent. Kosowski, Timmermann, Wermers, and White show that the tails of the distribution matter: averages hide skilled winners, and standard normal inference mishandles luck in a large cross section.
- **Economic importance.** The authors estimate that skilled right-tail funds generate about \$1.2 billion per year in value added after expenses and trading costs, while truly underperforming left-tail funds destroy about \$1.5 billion per year.
- **Takeaway.** Mutual fund performance is not simply a world of zero skill plus random luck. Skill exists, but it is concentrated in the right tail, most visible among growth funds, and must be separated from luck with a cross-sectional bootstrap rather than a simple fund-by-fund t-test.

2 Incremental contribution of the paper

- **From average performance to tail performance.** Much of the prior literature focuses on the average mutual fund. This paper asks a different question: whether the extreme right tail contains genuinely skilled managers after accounting for luck.
- **Formal treatment of luck.** The paper explicitly asks how many funds should have extreme alphas purely by sampling variation. This is crucial because, with many funds, some funds will look outstanding even when no manager has skill.
- **Cross-sectional bootstrap.** Instead of assuming a parametric normal distribution for ranked alphas, the paper constructs the empirical distribution of ranked alphas and ranked alpha t-statistics under the null of no skill.
- **Nonnormality and heterogeneous risk.** The paper emphasizes that mutual fund alphas have complicated distributions because individual fund residuals are often nonnormal and because funds take different levels of idiosyncratic risk.

- **t-statistic ranking.** Ranking funds by alpha can overweight short-lived or high-risk funds. Ranking by the alpha t-statistic normalizes alpha by its standard error and helps address heterogeneous risk-taking and fund lifespan differences.
- **Net-of-cost and pre-cost evidence.** The paper studies both net returns and stockholdings-based characteristic selectivity measures. This distinguishes skill in stock selection from costs, expenses, and trading frictions.
- **Persistence with bootstrap inference.** The persistence tests revisit Carhart-style performance persistence but replace standard parametric inference with bootstrapped inference. The result is stronger evidence of persistence among past winners than in Carhart's original interpretation.

3 Data and measurement

3.1 Mutual fund sample

The paper studies U.S. open-end domestic equity mutual funds from 1975 through 2002. Fund return data come from CRSP mutual fund files, and investment-objective information is supplemented with CDA/Thomson data.

- **Full sample.** The final database contains 2,118 domestic equity funds that exist for at least part of the 1975–2002 period.
- **Baseline bootstrap sample.** The baseline tests require at least 60 monthly net return observations, leaving 1,788 funds.
- **Investment objectives.** Funds are classified as aggressive growth, growth, growth-and-income, and balanced or income funds.
- **Share classes.** Fund-level returns are constructed by weighting share-class returns by beginning-of-month share-class assets.
- **Survival bias.** CRSP and CDA include both surviving and nonsurviving funds, so the merged database is designed to avoid the classic survivorship-bias problem.

Interpretation. The dataset is large enough for the cross-sectional problem to be real: if thousands of fund histories are examined ex post, ordinary single-fund inference will exaggerate the evidence for skill.

3.2 Benchmark models and alpha

The main performance model is the Carhart four-factor model:

$$r_{i,t} = \alpha_i + \beta_i RMRF_t + s_i SMB_t + h_i HML_t + p_i PR1YR_t + \varepsilon_{i,t},$$

where $r_{i,t}$ is fund i's excess net return, $RMRF$ is the market excess return, SMB is the size factor, HML is the book-to-market factor, and $PR1YR$ is the one-year momentum factor.

The paper also studies a conditional four-factor model that allows the market beta to vary with lagged public information, following Ferson and Schadt-style conditional performance evaluation.

- **Alpha.** α_i measures abnormal performance relative to the benchmark model.
- **Alpha t-statistic.** $t(\alpha_i)$ measures alpha scaled by its estimated standard error.
- **Main ranking variables.** The paper ranks funds both by alpha and by the alpha t-statistic.

Interpretation. The alpha ranking focuses on economic magnitude. The t-statistic ranking focuses on statistical reliability and is less distorted by high-risk or short-lived funds.

3.3 Stockholdings-based performance

The robustness tests also use the Daniel, Grinblatt, Titman, and Wermers characteristic selectivity measure, denoted CS:

$$CS_t = \sum_{j=1}^{N} w_{j,t-1} \left(R_{j,t} - R^b_{j,t} \right),$$

where each stock held by a fund is compared with a benchmark portfolio matched on size, book-to-market, and momentum.

- CS is closer to a pre-cost measure of stock-picking ability.
- The measure helps address concerns that Carhart alpha may misbenchmark certain nonlinear stock characteristics.
- Comparing CS with net-return alpha helps separate stock-picking skill from expenses and trading costs.

3.4 Economic value added

The paper computes skill-based value added by comparing actual funds above a given alpha threshold with the number of funds expected above that threshold by luck in the bootstrap.

- If 29 funds exceed 10 percent annual alpha but only 9 are expected by luck, the extra 20 funds are interpreted as the skill-related excess count at that threshold.
- Value added is approximated by multiplying alpha thresholds by fund assets and subtracting the bootstrap-implied lucky component.
- This converts statistical tail evidence into an estimate of investor wealth creation or destruction.

4 Methodology and empirical specifications

4.1 Why a bootstrap is needed

The paper argues that standard normal critical values are inappropriate for ranked mutual fund alphas except in a very special case: normally distributed residuals, no cross-fund residual correlation, identical risk levels, and no parameter-estimation complications.

In the real mutual fund data, several forces break this simple case.

- Individual funds often have nonnormal residuals because portfolios hold concentrated or nonlinear exposures.
- Funds differ sharply in idiosyncratic risk-taking and in sample length.
- Ex post sorting creates a multiple-comparisons problem: the top fund is not a randomly selected fund.
- The cross section of ranked alphas is a mixture of many fund-level alpha distributions.

Interpretation. The bootstrap is not just a technical refinement. It defines the relevant luck distribution for the ranked fund one actually observes.

4.2 Baseline residual-resampling bootstrap

The baseline bootstrap proceeds as follows.

1. Estimate the Carhart model for each fund and save estimated factor loadings and residuals.
2. Resample each fund's residuals with replacement.
3. Construct pseudo-returns using the estimated factor loadings and the resampled residuals, while imposing $\alpha_i = 0$.
4. Reestimate alpha and the alpha t-statistic on the pseudo-return series.
5. Sort funds in the bootstrap sample by alpha or by $t(\alpha)$.
6. Repeat the procedure 1,000 times to build the null distribution of the bottom fund, top fund, percentile-ranked funds, and other order statistics.

Interpretation. The resulting distribution answers the central question: how good should the best-looking managers look if every manager's true alpha is zero?

4.3 Alpha ranking versus t-statistic ranking

The paper uses two related but distinct tests.

- **Alpha ranking.** This directly asks whether funds with large economic alphas exceed what luck would produce.
- **t-statistic ranking.** This asks whether funds have statistically reliable abnormal performance after accounting for the precision of the alpha estimate.
- **Why the t-statistic matters.** Funds with short lives or high residual volatility can produce very large alpha estimates by chance. Scaling by the standard error reduces this problem.

4.4 Bootstrap extensions and robustness

The paper tests whether the baseline findings depend on the details of the bootstrap.

- **Stationary bootstrap.** Allows time-series dependence in residuals by resampling blocks.

- **Residual and factor resampling.** Resamples both residuals and factor returns.
- **Cross-sectional bootstrap.** Preserves potential cross-fund residual correlation by drawing common dates across funds.
- **Portfolio tests.** Tests whether groups of tail funds jointly have abnormal performance.
- **Minimum-history tests.** Repeats the analysis with minimum histories of 18, 30, 60, 90, and 120 months.
- **Omitted-factor checks.** Examines whether a missing persistent factor could explain extreme alphas.
- **Stockholdings tests.** Uses DGTW characteristic selectivity measures as a pre-cost performance benchmark.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether the observed right tail of mutual fund performance is too extreme to be explained by luck under a no-skill null, given the observed residual behavior, risk-taking, and fund histories. This is a statistical inference about the tail of the fund-performance distribution, not a randomized experiment assigning skill to managers.

5.2 Luck versus skill

A top fund can have a large alpha for two reasons: genuine stock-picking ability or lucky residual draws. The bootstrap simulates lucky residual draws while preserving key features of each fund's return process. If the actual ranked alpha is more extreme than the bootstrap distribution, the paper interprets this as evidence against pure luck.

5.3 Why the result differs from standard tests

A standard one-fund t-test asks whether one fixed fund has alpha different from zero. The paper's cross-sectional bootstrap asks whether the fund at a given rank in the sorted cross section has alpha or a t-statistic too extreme for the no-skill world.

- The bootstrap sometimes makes inference more conservative because some impressive-looking right-tail funds are plausible lucky outcomes.
- The bootstrap sometimes strengthens inference because the empirical ranked distribution can have thinner tails than a normal benchmark at some percentile points.
- The direction of the correction depends on the shape of the full cross-sectional null distribution.

5.4 Net performance and costs

The main alpha tests use net fund returns, so the skill evidence means that some managers cover expenses and trading costs. The stockholdings-based tests show broader pre-cost skill, but also show that much of the underperformance of bad funds comes from costs rather than from managers deliberately picking losing stocks.

5.5 Limitations

- Investors do not know ex ante which future right-tail funds will be truly skilled rather than lucky.
- Fund alphas depend on the benchmark model; although the paper tests many alternatives, no benchmark is perfect.
- Loads and taxes are not the focus of the net-return alpha tests.
- The analysis treats fund histories as the unit of observation; manager turnover and fund strategy changes can complicate interpretation.
- The evidence is strongest for the 1975–2002 sample and may differ in later market environments.

6 Tables and figures

6.1 Table 1: Summary statistics for the mutual fund database

Read:

- Table 1 reports the number of funds, excess returns, and four-factor alphas by five-year subperiod and investment objective.
- The baseline full-period sample contains 2,118 funds, of which 1,788 have at least 60 monthly return observations.
- Fund counts rise sharply over time, especially among growth funds.
- Equal-weighted fund portfolios have modestly negative full-period net four-factor alphas, consistent with the traditional finding that the average fund does not beat benchmarks after costs.

Economic message: The average fund is not the point of the paper. The table establishes the large fund universe and the fund histories needed to study the tails of the performance distribution.

Table I

Summary Statistics for Mutual Fund Database

This table reports the number, returns, and performance of U.S. open-end equity funds in existence during the 1975 to 2002 period. Panel A reports results for all investment objectives, Panel B for aggressive growth funds, Panel C for growth funds, Panel D for growth and income funds, and Panel E for balanced or income funds (these two categories are combined). The first column in each panel of this table shows the number of funds in existence in each subperiod. The second column of each panel reports the number of funds in existence with at least 60 monthly net return observations during the subperiod. The row "1971–1975," for example, shows that during the 1971 to 1975 subperiod, there existed 322 funds, but only 231 of them had 60 monthly observations during the 5-year subperiod. The third column reports the excess return in percent per year (12 times the average monthly return) of an equally weighted portfolio of funds during the subperiod. The fourth column reports the excess return in percent per year for an equally weighted portfolio of funds with at least 60 monthly observations during the subperiod. Column five reports the unconditional four-factor model alpha in percent per year for the equally weighted portfolio of funds during the subperiod. Column six reports the four-factor alpha in percent per year for funds that had at least 60 monthly observations during the subperiod.

	Number of Funds		Excess Return of Equally Weighted Portfolio (Pct/Year)		4-Factor-Alpha of Equally Weighted Portfolio (Pct/Year)	
Time Periods	≥1 Obs.	≥60 Obs.	≥1 Obs.	≥60 Obs.	≥1 Obs.	≥60 Obs.
Panel A: All Investment Objectives						
1971–1975	322	231	−9.8	−9.5	−0.1	0.1
1976–1980	342	282	11.9	12.1	−1.0	−0.7
1981–1985	459	300	4.1	4.0	0.2	0.1
1986–1990	796	421	9.5	9.6	0.2	0.0
1991–1995	1428	681	4.7	4.9	0.0	0.2
1996–2000	1929	1115	15.1	15.0	−1.8	−1.8
1998–2002	1824	1304	5.9	6.1	−1.0	−0.6
1975–2002	2118	1788	6.8	7.0	−0.5	−0.4
Panel B: Aggressive Growth Funds						
1971–1975	84	65	−12.6	−12.0	−0.4	0.0
1976–1980	88	76	15.8	16.0	−1.6	−1.3
1981–1985	132	79	3.8	3.5	−0.1	−0.7
1986–1990	208	122	9.8	9.8	0.5	0.1
1991–1995	240	177	7.1	7.4	1.0	1.2
1996–2000	242	191	17.4	17.3	−2.4	−2.3
1998–2002	232	200	6.6	7.0	−1.1	−0.8
1975–2002	285	264	8.0	8.0	−0.4	−0.4

Table 1, part 1: Summary statistics for all funds and aggressive growth funds.

Table I—*Continued*

	Number of Funds		Excess Return of Equally Weighted Portfolio (Pct/Year)		4-Factor-Alpha of Equally Weighted Portfolio (Pct/Year)	
Time Periods	≥1 Obs.	≥60 Obs.	≥1 Obs.	≥60 Obs.	≥1 Obs.	≥60 Obs.
Panel C: Growth Funds						
1971–1975	99	65	−11.2	−10.8	0.2	0.8
1976–1980	108	84	12.6	13.1	−1.2	−0.8
1981–1985	158	91	4.8	4.7	1.2	1.2
1986–1990	292	142	10.0	9.8	0.5	0.1
1991–1995	692	251	4.2	4.4	−0.5	−0.2
1996–2000	1145	527	16.1	16.1	−1.6	−1.7
1998–2002	1079	697	6.5	7.0	−0.7	−0.1
1975–2002	1227	985	7.2	7.4	−0.4	−0.1
Panel D: Growth and Income Funds						
1971–1975	89	64	−8.2	−7.8	−0.1	−0.1
1976–1980	93	76	10.3	10.7	−0.2	−0.1
1981–1985	112	82	4.0	3.8	0.0	−0.1
1986–1990	195	105	9.6	9.7	−0.4	−0.4
1991–1995	330	169	3.7	4.0	−0.2	0.0
1996–2000	355	265	13.6	13.8	−1.8	−1.6
1998–2002	335	270	4.6	4.7	−1.4	−1.2
1975–2002	396	353	6.2	6.2	−1.1	−1.0
Panel E: Balanced or Income Funds						
1971–1975	50	37	−5.6	−5.9	−0.6	−0.7
1976–1980	53	46	6.8	6.7	−0.8	−1.0
1981–1985	57	48	3.8	3.7	−0.4	−0.6
1986–1990	101	52	7.9	7.8	0.5	0.0
1991–1995	166	84	3.4	3.4	−0.2	−0.2
1996–2000	187	132	10.1	10.2	−1.9	−1.9
1998–2002	178	137	3.2	3.5	−1.1	−0.8
1975–2002	210	186	4.9	4.9	−0.7	−0.6

Table 1, part 2: Summary statistics for growth, growth-and-income, and balanced/income funds.

6.2 Table 2 and Figure 1: Baseline bootstrap evidence on alpha outliers

Read Table 2:

- Table 2 ranks funds by four-factor alpha and by the alpha t-statistic.
- The top and bottom funds have very large estimated alphas, but the bootstrap asks whether those values are extreme relative to a no-skill sorted cross section.
- Right-tail funds generally show significant evidence of positive net performance, especially when ranked by t-statistics.
- Below-median funds show strongly negative net alphas, suggesting that many funds underperform low-cost benchmarks after costs.

Read Figure 1:

- Figure 1 plots bootstrapped alpha distributions at several left-tail and right-tail ranks.
- The actual alpha is shown as a vertical line against the bootstrap null distribution.
- The shape of the bootstrap distribution changes sharply across the cross section, illustrating why normal critical values are unreliable.

Economic message: Some stars are real, but not every impressive alpha is skill. The relevant benchmark is what the top-ranked fund would look like in a no-skill world.

Table II

The Cross Section of Mutual Fund Alphas

In Panel A of this table, all U.S. open-end domestic equity funds that have at least 60 monthly net return observations during the 1975 to 2002 period are ranked on their four-factor alpha. The first to third rows report results when funds are ranked on their unconditional four-factor alphas. The fourth to sixth rows report results when funds are ranked on their conditional four-factor alphas. In Panel A, the first and second rows report the OLS estimate of alphas (in percent per month) and the cross-sectionally bootstrapped p-values of the alphas. For comparison, the third row reports the p-values of the t-statistic of alphas based on standard critical values of the t-statistic. The fourth and fifth rows of Panel A report conditional alphas (in percent per month) and the cross-sectionally bootstrapped p-values of conditional alphas. The sixth row reports the p-values of the t-statistic of alphas, based on standard critical values of the t-statistic. In Panel B, all U.S. open-end domestic equity funds that have at least 60 monthly net return observations during the 1975 to 2002 period are ranked on the t-statistics of their four-factor alpha (first row). The second row of Panel B shows the cross-sectionally bootstrapped p-values of the t-statistics. For comparison, the third row shows the p-values of the t-statistic based on standard critical values. Rows four to six show results when funds are ranked based on the t-statistic of their conditional four-factor model alpha. Rows four and five report the t-statistics of the conditional alpha and the cross-sectionally bootstrapped p-value of the t-statistic. Row six shows the p-values of the t-statistic of the conditional alphas, based on standard critical values. In each panel, the first columns on the left (right) report results for funds with the five lowest (highest) alphas or t-statistics, followed by results for marginal funds at different percentiles in the left (right) tail of the distribution. The cross-sectionally bootstrapped p-value is based on the distribution of the best (worst) funds in 1,000 bootstrap resamples. The t-statistics of alpha are based on heteroskedasticity- and autocorrelation-consistent standard errors.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	30%	40%	Median	40%	30%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel A: Funds Ranked on Four-Factor Model Alphas																									
Unconditional Alpha (Pct/Month)	−3.6	−2.7	−2.0	−1.7	−1.5	−0.8	−0.6	−0.5	−0.4	−0.2	−0.2	−0.1	−0.1	0.0	0.1	0.1	0.3	0.4	0.6	1.0	1.3	1.4	1.5	1.6	4.2
Cross-sectionally Bootstrapped p-Value	0.04	<0.01	<0.01	<0.01	<0.01	0.02	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1.00	1.00	0.99	0.02	<0.01	<0.01	<0.01	0.02	0.06	0.08	0.19	0.02
Parametric (Standard) p-Value	0.04	<0.01	<0.01	<0.01	<0.01	<0.01	0.06	0.01	0.09	<0.01	0.40	0.36		0.50	0.31	0.29	0.12	0.03	0.03	0.01	0.01	<0.01	0.12	0.01	<0.01
Conditional Alpha (Pct/Month)	−4.8	−2.2	−2.0	−1.9	−1.8	−1.0	−0.6	−0.5	−0.4	−0.3	−0.2	−0.1	−0.1	0.0	0.0	0.1	0.3	0.4	0.6	1.1	1.4	1.5	1.6	1.8	4.2
Cross-sectionally Bootstrapped p-Value	<0.01	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1	1	0.88	<0.01	<0.01	<0.01	<0.01	0.01	0.02	0.05	0.09	0.01
Parametric (Standard) p-Value	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.05	0.03	0.02	0.12	0.37		0.58	0.40	0.13	0.25	0.09	0.02	0.02	0.15	<0.01	0.01	0.01	<0.01
Panel B: Funds Ranked on the t-Statistics of Four-Factor Model Alphas																									
t-Unconditional Alpha	−7.9	−6.1	−6.0	−4.8	−4.5	−3.6	−3.0	−2.4	−1.9	−1.4	−0.9	−0.6	−0.4	0.0	0.3	0.8	1.4	2.0	2.3	2.8	3.5	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1.00	1.00	1.00	0.25	<0.01	<0.01	<0.01	0.04	0.08	0.01	0.03	<0.01
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.17	0.26		0.49	0.37	0.23	0.09	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
t-Conditional Alpha	−8.5	−5.8	−5.8	−5.7	−5.2	−3.8	−3.2	−2.7	−2.1	−1.6	−1.1	−0.7	−0.4	−0.1	0.3	0.8	1.4	2.0	2.3	2.9	3.5	3.7	3.9	4.6	5.8
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1.00	1.00	1.00	0.04	<0.01	<0.01	<0.01	0.01	0.02	0.02	0.01	0.01
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.02	0.06	0.13	0.23		0.55	0.40	0.22	0.08	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

Table 2: Cross section of mutual fund alphas and alpha t-statistics.

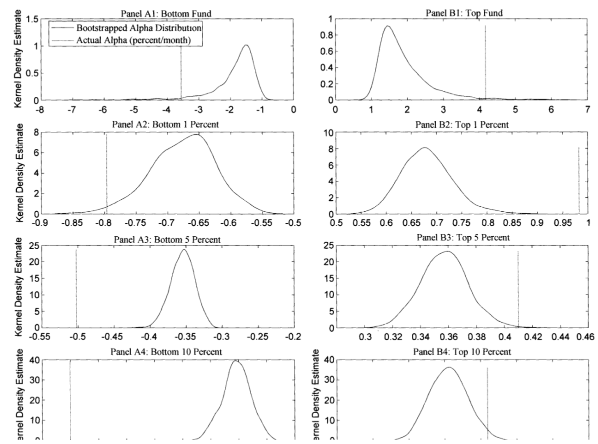


Figure 1: Actual alphas versus bootstrapped alpha distributions at selected ranks.

6.3 Figures 2 and 3: Nonnormality and economic value added

Read Figure 2:

- Figure 2 compares the actual cross-sectional distribution of alpha t-statistics with the bootstrapped null distribution.
- The actual distribution has more mass in the tails and a shape that is not well summarized by a normal approximation.
- This figure visually motivates the paper's reliance on bootstrap inference.

Read Figure 3:

- Figure 3 compares the actual number of funds above or below alpha thresholds with the number expected by bootstrap luck.
- At the 10 percent annual alpha threshold, 29 funds actually exceed the threshold, while only 9 are expected to do so by chance.
- The figure translates statistical evidence into wealth effects: skilled right-tail funds generate substantial value, while left-tail underperformers destroy value.

Economic message: The right tail contains economically meaningful skill, not just statistical curiosities. At the same time, poor funds also impose economically meaningful losses.

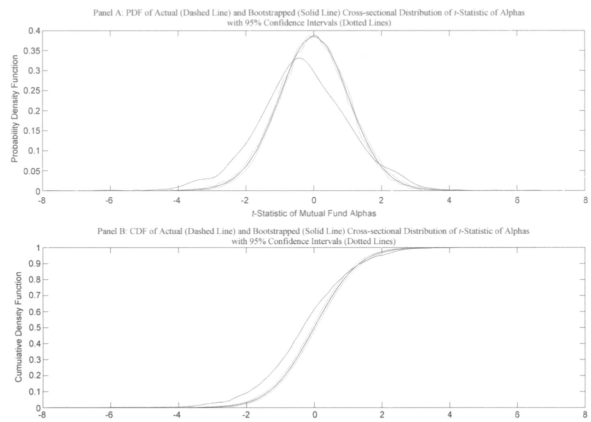


Figure 2: Actual versus bootstrapped cross-sectional distribution of alpha t-statistics.

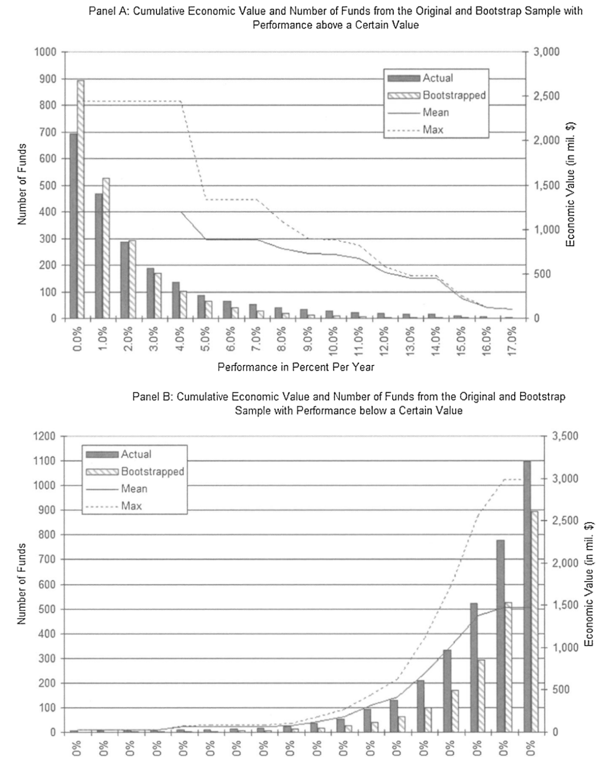


Figure 3: Cumulative fund counts and value added/destroyed above and below alpha thresholds.

6.4 Table 3: Subperiod evidence

Read:

- Table 3 repeats the baseline cross-sectional bootstrap separately for 1975–1989 and 1990–2002.
- The evidence suggests that outperforming managers became less common after 1990.
- Even in the later period, however, the top part of the performance distribution still shows evidence of genuine skill.
- Bootstrap inference differs more from standard inference in subperiod tests because samples are smaller and tails are more fragile.

Economic message: The market for mutual fund skill appears to change over time. Competition and market efficiency may reduce the prevalence of stars, but they do not eliminate them completely.

Table III

The Cross Section of Mutual Fund Alphas, for Subperiods

This table reports subperiod results for the cross section of funds' performance measures. In Panel A, all U.S. open-end domestic equity funds that have at least 60 monthly net return observations during the 1975 to 1989 period are ranked on their unconditional four-factor model alphas. The first and second rows report the OLS estimate of alphas (in percent per month) and the cross-sectionally bootstrapped p-values of alpha. For comparison, the third row reports the p-values of the t-statistic of alphas, based on standard critical values of the t-statistic. In Panel B, all U.S. open-end domestic equity funds that have at least 60 monthly net returns during the 1975 to 1989 period are ranked on the t-statistics of their unconditional four-factor model alpha. The first row shows the t-statistics of the unconditonal alpha. The second row reports the cross-sectionally bootstrapped p-values of the t-statistic of alphas. For comparison, the third row shows the p-values of the t-statistic, based on standard critical values. Panels C and D report the same measures as Panels A and B, but for the subperiod 1990 to 2002. In each panel, the first columns on the left (right) report results for funds with the five lowest (highest) alphas or t-statistics, followed by marginal funds at different percentiles in the left (right) tail of the distribution. The cross-sectionally bootstrapped p-value is based on the distribution of the best (worst) funds in 1,000 bootstrap resamples. The t-statistics of alpha are based on heteroskedasticity- and autocorrelation-consistent standard errors.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	30%	40%	Median	40%	30%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel A: Funds Ranked on Four-Factor Model Alphas (1975–1989)																									
Unconditional Alpha (Pct/Month)	−2.7	−1.4	−1.2	−1.1	−1.0	−1.0	−0.6	−0.5	−0.3	−0.2	−0.1	−0.1	0.0	0.1	0.1	0.2	0.3	0.5	0.6	0.7	0.7	0.8	0.9	1.0	1.0
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.03	0.10		<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.01	0.01	0.02	0.05	0.36
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.10	0.02	0.01	0.28	0.16	0.36		0.33	0.35	0.08	0.01	0.09	0.03	<0.01	<0.01	<0.01	0.01	<0.01	0.01
Panel B: Funds Ranked on t-Statistics of Four-Factor Model Alphas (1975–1989)																									
t-Unconditional Alpha	−6.1	−5.2	−4.7	−4.6	−4.0	−4.0	−2.9	−2.3	−1.8	−1.0	−0.6	−0.4	0.0	0.4	0.8	1.2	1.9	2.5	2.8	3.4	3.4	3.5	3.5	4.4	4.6
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.03	0.14	0.03		0.03	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	<0.01	0.02
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.16	0.27	0.35		0.36	0.21	0.11	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

Table 3, part 1: Subperiod bootstrap results for 1975–1989.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	30%	40%	Median	40%	30%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel C: Funds Ranked on Four-Factor Model Alphas (1990–2002)																									
Unconditional Alpha (Pct/Month)	−3.8	−3.6	−2.7	−1.5	−1.4	−0.9	−0.6	−0.5	−0.4	−0.3	−0.2	−0.1	−0.1	0.0	0.1	0.1	0.3	0.4	0.6	1.2	1.4	1.5	1.6	1.6	4.2
Cross-sectionally Bootstrapped p-Value	0.04	<0.01	<0.01	0.02	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1.00	1.00	0.99	0.23	<0.01	<0.01	<0.01	0.02	0.01	0.07	0.19	0.02
Parametric (Standard) p-Value	<0.01	0.04	<0.01	<0.01	<0.01	0.01	0.01	0.04	0.05	0.07	0.24	0.25		0.49	0.44	0.07	<0.01	0.03	0.16	<0.01	<0.01	0.12	<0.01	0.01	<0.01
Panel D: Funds Ranked on t-Statistics of Four-Factor Model Alphas (1990–2002)																									
t-Unconditional Alpha	−7.3	−6.6	−6.0	−5.0	−4.7	−3.6	−2.9	−2.5	−1.9	−1.4	−1.0	−0.6	−0.4	−0.1	0.3	0.7	1.3	1.9	2.2	2.8	3.4	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped p-Value	0.02	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01		1.00	1.00	1.00	0.50	<0.01	<0.01	0.01	0.04	0.12	<0.01	0.03	<0.01
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.17	0.27		0.48	0.38	0.23	0.09	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

Table 3, part 2: Subperiod bootstrap results for 1990–2002.

6.5 Table 4 and Figure 4: Investment-objective subgroups

Read Table 4:

- Table 4 studies fund performance separately by investment objective.
- Growth and aggressive growth funds show stronger evidence of genuine right-tail performance.
- Growth-and-income and balanced/income funds have right tails that look much more consistent with luck under the bootstrap.
- Standard parametric tests would often overstate evidence of skill in smaller or lower-alpha objective groups.

Read Figure 4:

- Figure 4 shows the bootstrapped t-statistic distributions for right-tail funds by investment objective.
- The distributions are visibly nonnormal and differ across fund categories.
- The bootstrap is especially important for aggressive growth funds and income-oriented funds, where ordinary normal inference can be misleading.

Economic message: Fund skill is not evenly distributed across objectives. The main evidence of skilled stock-picking is concentrated among growth-oriented funds.

Table IV

The Cross Section of Mutual Fund Alphas, by Investment Objective

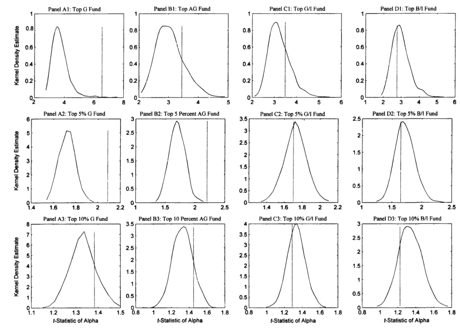
This table reports the cross section of alphas by investment objective. In Panel A all growth funds that have at least 60 monthly net return observations during the 1975 to 2002 period are ranked on their unconditional four-factor model alphas. The first and second rows report the OLS estimate of alphas (in percent per month) and the cross-sectionally bootstrapped p-value of the unconditional four-factor alpha. For comparison, the third row reports the p-values of the t-statistic of alphas, based on standard critical values of the t-statistic. In rows four to six of Panel A all growth funds that have at least 60 monthly net return observations during the 1975 to 2002 period are ranked on the t-statistics of their unconditional four-factor model alphas. The fourth row shows the t-statistics of the alpha. The fifth row reports the cross-sectionally bootstrapped p-values of the t-statistic of alphas. For comparison, the sixth row shows the p-values of the t-statistic, based on standard critical values. Panels B, C, and D report the same measures as Panel A, but for the investment objectives of aggressive growth, growth and income, and balanced or income. In each panel, the first columns on the left (right) report results for funds with the five lowest (highest) alphas or t-statistics, followed by results for marginal funds at different percentiles in the left (right) tail of the distribution. The cross-sectionally bootstrapped p-value is based on the distribution of the best (worst) funds in 1,000 bootstrap resamples. The t-statistics of alpha are based on heteroskedasticity- and autocorrelation-consistent standard errors.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel A: Growth Funds Ranked on Four-Factor Model Alphas																				
Unconditional Alpha (Pct/Month)	−3.6	−2.7	−1.7	−1.5	−1.2	−0.8	−0.6	−0.5	−0.4	−0.3	0.1	0.3	0.4	0.6	1.0	1.3	1.4	1.5	1.6	4.2
Cross-sectionally Bootstrapped p-Value	0.05	<0.01	0.01	0.01	0.07	0.25	0.02	<0.01	<0.01	<0.01	0.99	0.02	<0.01	<0.01	<0.01	0.02	0.06	0.08	0.19	0.02
Parametric (Standard) p-Value	0.04	<0.01	<0.01	<0.01	<0.01	0.06	<0.01	0.18	0.08	0.31	0.29	0.12	0.03	0.03	0.01	0.01	0.00	0.12	0.01	<0.01
Growth Funds Ranked on t-Statistics of Four-Factor Model Alphas																				
t-Unconditional Alpha	−7.9	−4.8	−4.2	−4.0	−3.9	−3.6	−2.9	−2.4	−1.8	−1.3	0.8	1.4	2.0	2.3	2.8	3.5	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	1.00	0.25	<0.01	<0.01	<0.01	0.04	0.08	0.01	0.03	<0.01
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.04	0.10	0.23	0.09	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel B: Aggressive Growth Funds Ranked on Four-Factor Model Alphas																				
Unconditional Alpha (Pct/Month)	−2.0	−1.2	−1.1	−1.1	−0.9	−1.1	−0.7	−0.6	−0.4	−0.3	0.2	0.3	0.6	0.7	0.9	0.9	0.9	0.9	0.9	1.1
Cross-sectionally Bootstrapped p-Value	<0.01	0.02	0.01	<0.01	<0.01	0.01	<0.01	<0.01	<0.01	<0.01	0.33	<0.01	<0.01	<0.01	0.05	<0.01	0.02	0.05	0.20	0.37
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.20	0.07	0.01	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Aggressive Growth Funds Ranked on t-Statistics of Four-Factor Model Alphas																				
t-Unconditional Alpha	−6.1	−3.8	−1.5	−2.1	−2.4	−1.5	−4.5	−1.7	−1.5	−1.2	0.7	2.4	2.6	2.9	2.2	3.1	1.2	2.2	2.3	3.5
Cross-sectionally Bootstrapped p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.63	0.13	<0.01	<0.01	0.04	0.02	0.06	0.04	0.12	0.19
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.20	0.07	0.01	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

Table 4, part 1: Growth and aggressive growth fund bootstrap results.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel C: Growth and Income Funds Ranked on Four-Factor Model Alphas																				
Unconditional Alpha (Pct/Month)	−0.9	−0.8	−0.8	−0.7	−0.6	−0.7	−0.5	−0.4	−0.3	−0.2	0.1	0.1	0.3	0.3	0.5	0.4	0.5	0.5	0.6	0.8
Cross-sectionally Bootstrapped p-Value	0.11	0.03	0.01	0.01	<0.01	0.01	<0.01	<0.01	<0.01	<0.01	0.97	1.00	0.22	0.35	0.27	0.35	0.27	0.20	0.39	0.26
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.03	0.06	0.15	0.24	0.31	0.17	0.01	0.09	0.02	0.03	0.02	0.09	0.03	0.05
Growth and Income Funds Ranked on t-Statistics of Four-Factor Model Alphas																				
t-Unconditional Alpha	−4.4	−4.0	−3.8	−3.6	−3.6	−3.6	−3.0	−2.5	−2.1	−1.4	0.7	1.3	1.7	1.9	2.5	2.4	2.5	2.6	2.7	3.5
Cross-sectionally Bootstrapped p-Value	0.03	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.99	0.68	0.59	0.69	0.46	0.35	0.46	0.49	0.63	0.27
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.02	0.08	0.25	0.10	0.05	0.03	0.01	0.01	0.01	0.01	<0.01	<0.01
Panel D: Balanced or Income Funds Ranked on Four-Factor Model Alphas																				
Unconditional Alpha (Pct/Month)	−0.6	−0.6	−0.5	−0.4	−0.4	−0.6	−0.3	−0.3	−0.3	−0.2	0.1	0.1	0.2	0.3	0.4	0.3	0.3	0.4	0.4	0.5
Cross-sectionally Bootstrapped p-Value	0.37	0.05	0.01	0.06	0.03	0.05	0.01	<0.01	<0.01	<0.01	0.95	0.40	0.46	0.22	0.37	0.17	0.35	0.17	0.37	0.47
Parametric (Standard) p-Value	0.01	0.01	0.02	0.15	0.02	0.01	0.04	0.05	<0.01	0.04	0.17	0.21	0.05	0.11	<0.01	0.10	0.09	0.01	<0.01	0.05
Balanced or Income Funds Ranked on t-Statistics of Four-Factor Model Alphas																				
t-Unconditional Alpha	−6.0	−3.6	−3.3	−3.3	−3.2	−3.6	−3.1	−2.6	−2.3	−1.6	0.7	1.2	1.7	2.0	2.4	2.1	2.1	2.2	2.4	2.8
Cross-sectionally Bootstrapped p-Value	<0.01	0.01	<0.01	<0.01	<0.01	0.01	<0.01	<0.01	<0.01	<0.01	0.98	0.81	0.65	0.51	0.71	0.50	0.69	0.74	0.71	0.64
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.01	0.05	0.26	0.11	0.05	0.03	0.01	0.02	0.02	0.02	0.01	<0.01

Table 4, part 2: Growth-and-income and balanced/income fund bootstrap results.



igure 4. Estimated alpha *t*-statistics vs. bootstrapped alpha *t*-statistic distributions for individual funds at various percentile points the cross section, by investment objective. This figure plots kernel density estimates of the bootstrapped distribution of the *t*-statistic of alpha olid line) for U.S. equity funds with at least 60 monthly net return observations during the 1975 to 2002 period. Panels A1–A3 show results for owth funds (G), Panels B1–B3 for aggressive growth funds (AG), Panels C1–C3 for growth and income funds (G/I), and Panels D1–D3 for balanced income funds (B/I). The funds are marginal funds in the right tail of the distribution. The *x*-axis shows the *t*-statistic, and the *y*-axis shows the rnel density. The dashed vertical line represents the actual *t*-statistic of alpha. The results are based on the unconditional four-factor model.

Figure 4: Bootstrapped alpha t-statistic distributions by investment objective.

Table V

Sensitivity Analysis of the Cross Section of Mutual Fund Alphas to Minimum Number of Observations

In Panel A, all U.S. open-end domestic equity funds that have at least 18 monthly net return observations during the 1975 to 2002 period are ranked on the *t*-statistics of their nconditional four-factor model alphas. The first row shows the *t*-statistics of the unconditional alphas. The second row reports the cross-sectionally bootstrapped *p*-values of the -statistics. For comparison, the third row shows the *p*-value of the *t*-statistic, based on standard critical values. Panels B, C, D, and E report the same measures as Panel A, but for funds with a minimum number of observations of 30, 60, 90, and 120 monthly net return observations, respectively. In each panel, the first columns on the left (right) report results for funds with the five lowest (highest) *t*-statistics, followed by results for marginal funds at different percentiles in the left (right) tail of the distribution. The cross-sectionally bootstrapped -value is based on the distribution of the best (worst) funds in 1,000 bootstrap resamples. The *t*-statistics of alpha are based on heteroskedasticity- and autocorrelation-consistent standard errors.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel A: Minimum of 18 Observations Per Fund																				
-Unconditional Alpha	−7.9	−6.1	−6.0	−4.8	−4.5	−3.8	−3.1	−2.5	−2.1	−1.4	0.7	1.3	1.9	2.2	2.7	3.5	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped *p*-Value	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	1.00	0.93	0.01	<0.01	0.04	0.20	0.34	0.06	0.15	0.03
Parametric (Standard) *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.02	0.08	0.25	0.10	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel B: Minimum of 30 Observations Per Fund																				
-Unconditional Alpha	−7.9	−6.1	−6.0	−4.8	−4.5	−3.7	−3.1	−2.5	−2.1	−1.4	0.7	1.3	1.9	2.3	2.7	3.5	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	1.00	0.83	<0.01	<0.01	0.02	0.11	0.20	0.03	0.09	0.01
Parametric (Standard) *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.02	0.08	0.24	0.10	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel C: Minimum of 60 Observations Per Fund																				
-Unconditional Alpha	−7.9	−6.1	−6.0	−4.8	−4.5	−3.6	−3.0	−2.4	−1.9	−1.4	0.8	1.4	2.0	2.3	2.8	3.5	3.5	4.1	4.2	6.6
Cross-sectionally Bootstrapped *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	1.00	0.25	<0.01	<0.01	<0.01	0.04	0.08	0.01	0.03	<0.01
Parametric (Standard) *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.23	0.09	0.03	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel D: Minimum of 90 Observations Per Fund																				
-Unconditional Alpha	−7.9	−6.1	−6.0	−4.8	−4.5	−3.6	−2.6	−2.4	−1.9	−1.4	0.8	1.4	1.9	2.2	2.6	3.1	3.2	3.4	3.5	3.5
Cross-sectionally Bootstrapped *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.97	0.06	0.01	<0.01	0.13	0.13	0.16	0.11	0.30	0.64
Parametric (Standard) *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.09	0.22	0.08	0.03	0.01	0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel E: Minimum of 120 Observations Per Fund																				
-Unconditional Alpha	−7.9	−6.1	−4.8	−4.5	−3.8	−3.6	−2.6	−2.4	−2.0	−1.3	0.9	1.5	2.0	2.3	2.7	3.1	3.2	3.4	3.5	3.5
Cross-sectionally Bootstrapped *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.46	0.01	<0.01	<0.01	0.01	0.02	0.02	0.02	0.12	0.40
Parametric (Standard) *p*-Value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.03	0.10	0.19	0.07	0.02	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

Table 5: Sensitivity of alpha t-statistic results to the minimum return-history requirement.

6.6 Table 5: Sensitivity to return-history requirements

Read:

- Table 5 repeats the t-statistic bootstrap using minimum fund histories of 18, 30, 60, 90, and 120 months.
- The main conclusions are qualitatively robust across these inclusion rules.
- The 120-month requirement eliminates many extreme short-history funds, but the bootstrap then moves deeper into the distribution to evaluate the remaining right tail.

Economic message: The evidence is not driven by one arbitrary minimum-history cutoff. The bootstrap adapts to changes in the included fund universe.

6.7 Table 6: Stockholdings-based characteristic selectivity

Read:

- Table 6 uses the DGTW characteristic selectivity measure, which evaluates stocks held by funds against matched benchmark portfolios.
- This is closer to a pre-cost measure of stock-picking ability.
- Right-tail evidence is broadly consistent with the net-alpha results and, as expected, extends farther into the distribution before costs are deducted.
- The table finds little evidence of gross underperformance. The poor net performance of bad funds is mainly due to expenses and trading costs rather than negative stock-selection ability.

Economic message: Some managers can pick stocks, but costs matter. Positive pre-cost ability does not automatically translate into positive net investor returns.

Table VI

The Cross Section of Stockholdings-Based Performance Measures

In Panel A, all U.S. open-end domestic equity funds that have holdings available for at least 5 years during the 1975 to 2002 period are ranked on the CS measure introduced by Daniel et al. (1997). The first and second rows of Panel A report the CS measures (in percent per month) and the cross-sectionally bootstrapped p-values of the CS measures. For comparison, the third row reports the p-values of the t-statistic of the CS measures based on standard critical values of the t-statistic. In rows four to six funds are ranked on the t-statistics of their CS measures. The fourth row shows the t-statistics. The fifth row reports the cross-sectionally bootstrapped p-values of the t-statistic. For comparison, the sixth row shows the p-values of the t-statistic, based on standard critical values. Panels B, C, D, and E report the same measures as Panel A but for growth, aggressive growth, growth and income, and balanced or income funds, respectively. In each panel, the first columns on the left (right) report results for funds with the five lowest (highest) CS measures or t-statistics, followed by results for marginal funds at different percentiles in the left (right) tail of the distribution. The cross-sectionally bootstrapped p-value is based on the distribution of the best (worst) funds in 1,000 bootstrap resamples. The t-statistics of the CS measure are based on heteroskedasticity- and autocorrelation-consistent standard errors.

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel A: All Investment Objectives – Funds Ranked on CS Measure																				
CS (Pct/Month)	−2.4	−2.0	−1.6	−1.5	−1.5	−0.7	−0.4	−0.3	−0.2	−0.1	0.2	0.4	0.6	0.7	0.9	1.2	1.3	1.4	1.6	1.7
Cross-sectionally Bootstrapped p-Value	0.29	0.14	0.16	0.09	0.05	0.76	1.00	1.00	1.00	1.00	<0.01	<0.01	<0.01	<0.01	0.19	0.29	0.35	0.40	0.51	0.74
Parametric (Standard) p-Value	0.03	<0.01	0.16	<0.01	0.09	0.07	0.06	0.19	0.12	0.14	0.17	0.08	0.01	0.03	<0.01	<0.01	0.21	0.05	0.05	0.07
Funds Ranked on t-Statistics of CS Measure																				
t-Unconditional CS	−4.3	−3.2	−3.1	−3.0	−3.0	−2.1	−1.6	−1.4	−1.0	−0.5	1.2	1.7	2.0	2.3	2.9	3.1	3.2	3.3	3.5	3.7
Cross-sectionally Bootstrapped p-Value	0.13	0.76	0.60	0.49	0.49	1.00	1.00	1.00	1.00	1.00	<0.01	<0.01	<0.01	<0.01	<0.01	0.07	0.09	0.18	0.21	0.34
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	0.02	0.06	0.09	0.16	0.32	0.12	0.05	0.02	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Panel B: Growth Funds Ranked on CS Measure																				
CS (Pct/Month)	−2.4	−2.0	−1.5	−1.0	−1.0	−0.8	−0.5	−0.4	−0.2	−0.1	0.3	0.4	0.6	0.7	0.9	1.1	1.2	1.4	1.6	1.7
Cross-sectionally Bootstrapped p-Value	0.14	0.05	0.11	0.78	0.65	0.90	0.94	1.00	1.00	1.00	<0.01	<0.01	<0.01	<0.01	0.25	0.25	0.26	0.21	0.31	0.57
Parametric (Standard) p-Value	0.03	<0.01	<0.01	0.18	0.16	0.30	0.03	0.20	0.17	0.22	0.28	0.18	0.03	0.04	0.03	0.01	<0.01	0.05	0.05	0.07
Growth Funds Ranked on t-Statistics of CS Measure																				
t-Unconditional CS	−4.3	−3.2	−3.0	−3.0	−2.8	−2.4	−1.7	−1.4	−1.0	−0.5	1.3	1.7	2.0	2.2	2.9	3.1	3.1	3.2	3.3	3.5
Cross-sectionally Bootstrapped p-Value	0.11	0.51	0.40	0.31	0.29	0.78	1.00	1.00	1.00	1.00	<0.01	<0.01	<0.01	0.01	<0.01	0.01	0.05	0.11	0.25	0.42
Parametric (Standard) p-Value	<0.01	<0.01	<0.01	<0.01	<0.01	0.01	0.05	0.08	0.17	0.31	0.10	0.05	0.02	0.02	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

(continued)

Table VI—*Continued*

	Bottom	2.	3.	4.	5.	1%	3%	5%	10%	20%	20%	10%	5%	3%	1%	5.	4.	3.	2.	Top
Panel C: Aggressive Growth Funds Ranked on CS Measure																				
CS (Pct/Month)	−1.5	−0.9	−0.7	−0.6	−0.4	−0.7	−0.4	−0.3	−0.2	−0.1	0.2	0.4	0.5	0.7	0.9	0.8	0.9	0.9	0.9	1.0
Cross-sectionally Bootstrapped p-Value	0.2	0.32	0.53	0.82	0.99	0.53	0.97	1.00	1.00	1.00	<0.01	<0.01	<0.01	0.01	0.17	0.06	0.05	0.17	0.39	0.68
Parametric (Standard) p-Value	0.09	0.05	0.04	0.20	0.08	0.04	0.06	0.10	0.29	0.40	0.01	0.02	0.09	0.07	0.07	0.01	<0.01	0.07	<0.01	0.07
Aggressive Growth Funds Ranked on t-Statistics of CS Measure																				
t-Unconditional CS	−1.8	−1.6	−1.6	−1.6	−1.4	−1.6	−1.3	−1.2	−0.8	−0.4	1.2	1.7	2.1	2.3	2.8	2.5	2.7	2.8	2.9	3.7
Cross-sectionally Bootstrapped p-Value	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	<0.01	<0.01	<0.01	0.02	0.04	0.07	0.04	0.04	0.13	0.05
Parametric (Standard) p-Value	0.04	0.05	0.05	0.06	0.08	0.05	0.09	0.12	0.21	0.34	0.12	0.04	0.02	0.01	<0.01	0.01	<0.01	<0.01	<0.01	<0.01
Panel D: Growth and Income Funds Ranked on CS Measure																				
CS (Pct/Month)	−1.6	−0.9	−0.4	−0.4	−0.3	−0.4	−0.3	−0.2	−0.2	−0.1	0.1	0.2	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.7
Cross-sectionally Bootstrapped p-Value	0.15	0.07	0.92	0.92	0.92	0.92	0.98	0.91	0.89	1.00	0.08	0.18	<0.01	0.01	0.40	0.12	0.26	0.40	0.65	0.74
Parametric (Standard) p-Value	0.16	<0.01	<0.01	0.17	0.08	<0.01	0.10	0.03	0.22	0.16	0.15	0.06	0.08	0.01	0.06	0.01	0.06	0.06	0.01	0.04
Growth and Income Funds Ranked on t-Statistics of CS Measure																				
t-Unconditional CS	−3.1	−2.6	−2.6	−2.0	−2.0	−2.6	−1.6	−1.5	−1.1	−0.6	1.0	1.6	2.0	2.4	2.7	2.4	2.4	2.7	2.9	3.0
Cross-sectionally Bootstrapped p-Value	0.42	0.55	0.37	0.94	0.86	0.37	0.99	0.97	0.96	1.00	0.02	0.02	0.03	0.02	0.13	0.13	0.26	0.13	0.20	0.49
Parametric (Standard) p-Value	<0.01	<0.01	0.01	0.02	0.02	0.01	0.06	0.07	0.13	0.28	0.15	0.06	0.02	0.01	<0.01	0.01	0.01	<0.01	<0.01	<0.01
Panel E: Balanced or Income Funds Ranked on CS Measure																				
CS (Pct/Month)	−0.4	−0.3	−0.3	−0.2	−0.2	−0.4	−0.4	−0.3	−0.3	−0.1	0.2	0.3	0.4	1.3	1.3	0.2	0.2	0.3	0.4	1.3
Cross-sectionally Bootstrapped p-Value	0.69	0.42	0.34	0.49	0.40	0.69	0.69	0.42	0.34	0.79	0.13	0.27	0.39	0.22	0.22	0.13	0.25	0.27	0.39	0.22
Parametric (Standard) p-Value	0.23	0.10	0.01	0.17	0.21	0.23	0.23	0.10	0.01	0.24	0.05	0.10	0.03	0.21	0.21	0.17	0.05	0.10	0.03	0.21
Balanced and Income Funds Ranked on t-Statistics of CS Measure																				
t-Unconditional CS	−2.4	−1.3	−1.0	−0.8	−0.7	−2.4	−2.4	−1.3	−1.0	−0.7	1.1	1.7	1.9	2.4	2.4	1.3	1.7	1.7	1.9	2.44
Cross-sectionally Bootstrapped p-Value	0.32	0.81	0.88	0.86	0.79	0.32	0.32	0.81	0.88	0.69	0.17	0.18	0.23	0.23	0.23	0.16	0.05	0.18	0.23	0.23
Parametric (Standard) p-Value	0.01	0.1	0.17	0.21	0.23	0.01	0.01	0.10	0.17	0.24	0.14	0.05	0.03	0.01	0.01	0.10	0.05	0.05	0.03	0.01

Table 6, part 1: Stockholdings-based CS results for all funds and growth funds.

Table 6, part 2: Stockholdings-based CS results for other investment objectives.

6.8 Table 7: Bootstrap tests of performance persistence

Read:

- Table 7 ranks funds using past alpha and then evaluates future one-year performance using bootstrap inference.
- Panel A uses a three-year ranking period and a one-year holding period.
- The top-ranked fund and top decile show positive future alphas; the bootstrap provides stronger evidence that the top decile contains skilled funds than the standard parametric test.
- Panel B uses a one-year ranking period and shows that short-horizon ranked portfolios have more nonnormality, making bootstrap inference especially important.

Economic message: The paper overturns the strong version of the claim that winners do not persist. Superior net performance among top-ranked funds can persist, especially for growth-oriented funds, once inference accounts for nonnormality and ex post sorting.

Table VII

Bootstrap Performance Persistence Tests—All Investment Objectives

In Panel A, mutual funds are sorted on January 1 each year (from 1978 until 2002) into decile portolios based on their unconditional four-factor model alphas estimated over the prior 3 years. We require a minimum of 36 monthly net return observations for this estimate. For funds that have missing observations during these prior 3 years, observations from the 12 months preceding the 3-year window are added to obtain 36 observations. This assures that funds with missing observations are not excluded. The portfolios are equally weighted monthly, so the weights are readjusted whenever a fund disappears. Funds with the highest past 3-year return comprise decile 1, and funds with the lowest comprise decile 10. The "5%ile" portfolio is an equally weighted portfolio of the top5% funds. The last four rows represent the difference in returns between the top and bottom deciles (5 percentiles, 1 percentiles), as well as between the ninth and tenth deciles. In Panel B, the portfolios are formed based on past 1-year alphas, and funds are held for 1 year. Column five reports the one-tailed parametric p-value of alpha. Columns six and seven report the cross-sectionally bootstrapped p-values for the t-statistic of alpha. Column six reports the probability that the bootstrapped t-statistic of alpha is lower than $(-|t(\text{alpha})|)$, i.e., the left tail of the bootstrapped distribution. Column seven reports the probability that the bootstrapped t-statistic of alpha is higher than $(+|t(\text{alpha})|)$, i.e., the right tail of the bootstrapped distribution. Columns 12 and 13 report the adjusted R-squared and the annual expense ratio. The expense ratio is calculated as the time-series average of the cross-sectional average of the expense ratios of the funds in the portfolios. The last three columns report the skewness, kurtosis, and the p-value of the Jarque-Bera nonnormality statistics. RMRF, SMB, and HML are Fama and French's (1993) market proxy and factor-mimicking portfolios for size and book-to-market equity. PR1YR is a factor-mimicking portfolio for 1-year return momentum. Alpha (in percent per month) is the intercept of the model.

Panel A: 3-Year Ranking Periods, 1-Year Holding Period

Fractile	Excess Ret. (Pct/Month)	Std. Dev.	Alpha (Pct/Month)	t-Stat of Alpha	One-Tailed Parametric p-Value of Alpha	Bootstr. p-Value of t(Alpha) (Left Tail)	Bootstr. p-Value t(Alpha) (Right Tail)	RMRF	SMB	HML	PR1YR	Adj. R^2	Exp. Ratio	Skew.	Kur.	p-Value (JB-Test)
Top	1.04	7.64	0.48	1.5	0.06	0.05	0.04	0.91	0.46	−0.56	0.19	0.65	1.04	0.2	5.2	<0.01
1%ile	0.54	6.79	0.11	0.7	0.24	0.38	0.20	1.03	0.53	−0.45	−0.06	0.89	0.98	−0.2	4.6	<0.01
5%ile	0.58	5.65	0.12	1.3	0.09	0.13	0.03	0.97	0.40	−0.27	−0.04	0.95	1.01	−0.1	5.0	<0.01
1.Dec	0.57	5.17	0.08	1.1	0.13	0.23	0.05	0.95	0.33	−0.15	−0.04	0.96	0.97	−0.2	4.7	<0.01
2.Dec	0.53	4.42	0.02	0.3	0.39	0.60	0.23	0.90	0.15	−0.02	−0.01	0.97	0.87	0.3	4.6	<0.01
3.Dec	0.53	4.21	−0.02	−0.4	0.35	0.45	0.27	0.90	0.10	0.06	0.01	0.98	0.86	0.3	4.7	<0.01
4.Dec	0.50	4.06	−0.02	−0.3	0.38	0.57	0.15	0.88	0.08	0.06	−0.01	0.98	0.84	0.1	4.4	<0.01
5.Dec	0.49	4.13	−0.04	−1.0	0.16	0.32	0.04	0.90	0.05	0.08	−0.01	0.98	0.86	0.3	5.3	<0.01
6.Dec	0.46	4.04	−0.08	−2.3	0.01	0.03	<0.01	0.89	0.03	0.08	0.01	0.98	0.85	0.4	5.8	<0.01
7.Dec	0.48	4.12	−0.09	−1.9	0.03	0.03	<0.01	0.90	0.05	0.10	0.02	0.97	0.88	0.2	4.8	<0.01
8.Dec	0.52	4.08	−0.07	−1.7	0.05	0.07	0.01	0.89	0.10	0.11	0.04	0.97	0.94	0.3	4.7	<0.01
9.Dec	0.50	4.25	−0.09	−1.6	0.06	0.10	0.03	0.90	0.14	0.08	0.03	0.97	1.03	0.5	6.2	<0.01
10.Dec	0.33	4.54	−0.29	−4.2	<0.01	<0.01	<0.01	0.93	0.26	0.08	0.04	0.96	1.30	0.7	5.5	<0.01
95%ile	0.15	4.76	−0.49	−6.0	<0.01	<0.01	<0.01	0.95	0.31	0.06	0.04	0.95	1.50	0.3	4.3	<0.01
99%ile	−0.22	5.66	−0.89	−5.2	<0.01	<0.01	<0.01	1.04	0.40	0.09	0.00	0.81	2.47	1.0	8.2	<0.01
Bottom	−1.02	11.69	−1.38	−2.6	<0.01	<0.01	<0.01	0.87	0.98	−0.42	−0.11	0.32	5.26	2.4	20.5	<0.01
Sprd10%	0.24	1.57	0.37	3.8	<0.01	<0.01	<0.01	0.02	0.08	−0.23	−0.07	0.36	−0.32	−0.3	4.7	<0.01
Sprd5%	0.42	2.10	0.61	4.7	<0.01	<0.01	<0.01	0.01	0.09	−0.33	−0.08	0.36	−0.49	−0.4	4.4	<0.01
Sprd1%	0.76	3.89	1.00	4.0	<0.01	<0.01	<0.01	−0.01	0.14	−0.55	−0.06	0.25	−1.49	−0.8	5.7	<0.01
Dec9-10	0.17	0.85	0.20	4.5	<0.01	<0.01	<0.01	−0.03	−0.12	0.01	0.00	0.27	−0.27	0.0	6.5	<0.01

(continued)

Table VII—*Continued*

Panel B: 1-Year Ranking Periods, 1-Year Holding Period

Fractile	Excess Ret. (Pct/Month)	Std. Dev.	Alpha (Pct/Month)	t-Stat of Alpha	One-Tailed Parametric p-Value of Alpha	Bootstr. p-Value of t(Alpha) (Left Tail)	Bootstr. p-Value t(Alpha) (Right Tail)	RMRF	SMB	HML	PR1YR	Adj. R^2	Exp. Ratio	Skew.	Kur.	p-Value (JB-Test)
Top	0.84	9.14	0.14	0.4	0.36	0.39	0.31	1.10	0.85	−0.27	<0.01	0.55	1.06	0.3	8.7	<0.01
1%ile	0.79	6.50	0.14	0.9	0.19	0.13	0.24	1.04	0.61	−0.26	0.04	0.87	1.26	0.2	3.7	0.01
5%ile	0.82	5.69	0.14	1.5	0.06	0.05	0.06	1.00	0.49	−0.17	0.09	0.93	1.09	0.6	5.1	<0.01
1.Dec	0.78	5.30	0.14	1.9	0.03	0.04	0.01	0.97	0.41	−0.14	0.07	0.95	1.04	0.7	5.9	<0.01
2.Dec	0.66	4.49	0.07	1.5	0.07	0.11	<0.01	0.92	0.20	−0.01	0.03	0.97	0.92	0.4	4.9	<0.01
3.Dec	0.58	4.21	0.03	0.6	0.27	0.53	0.08	0.90	0.12	0.04	0.01	0.97	0.88	0.6	5.4	<0.01
4.Dec	0.54	4.08	<0.01	−0.1	0.46	0.80	0.15	0.89	0.09	0.06	<0.01	0.98	0.86	0.3	5.7	<0.01
5.Dec	0.53	4.01	−0.01	−0.3	0.37	0.62	0.17	0.88	0.07	0.07	<0.01	0.98	0.85	0.5	5.0	<0.01
6.Dec	0.46	3.97	−0.06	−1.8	0.04	0.10	<0.01	0.88	0.06	0.08	<0.01	0.98	0.87	0.2	4.0	<0.01
7.Dec	0.45	4.00	−0.08	−2.0	0.02	0.08	<0.01	0.88	0.06	0.07	<0.01	0.98	0.90	0.3	5.0	<0.01
8.Dec	0.42	4.12	−0.15	−3.1	<0.01	0.01	<0.01	0.90	0.09	0.07	0.02	0.97	0.92	0.0	4.1	<0.01
9.Dec	0.40	4.17	−0.18	−3.2	<0.01	<0.01	<0.01	0.91	0.11	0.09	0.01	0.96	0.95	−0.1	4.2	<0.01
10.Dec	0.28	4.39	−0.30	−3.7	<0.01	<0.01	<0.01	0.92	0.23	0.10	−0.03	0.93	1.22	−0.1	4.5	<0.01
95%ile	0.22	4.57	−0.39	−4.5	<0.01	<0.01	<0.01	0.94	0.28	0.11	−0.03	0.92	1.35	−0.1	4.1	<0.01
99%ile	−0.03	4.96	−0.66	−5.1	<0.01	<0.01	<0.01	0.93	0.41	0.14	−0.04	0.82	1.87	−0.2	4.5	<0.01
Bottom	−0.59	9.96	−1.31	−2.7	<0.01	<0.01	<0.01	1.12	0.25	0.50	−0.17	0.23	2.20	2.8	24.8	<0.01
Sprd10%	0.51	2.22	0.44	3.9	<0.01	<0.01	<0.01	0.05	0.18	−0.24	0.10	0.39	−0.18	0.4	4.5	<0.01
Sprd5%	0.61	2.67	0.53	4.0	<0.01	<0.01	<0.01	0.06	0.21	−0.28	0.12	0.36	−0.26	0.5	4.3	<0.01
Sprd1%	0.82	3.82	0.80	3.9	<0.01	<0.01	<0.01	0.11	0.21	−0.40	0.08	0.28	−0.60	0.2	3.1	0.34
Dec9-10	0.12	0.86	0.12	2.2	0.01	0.01	<0.01	−0.01	−0.12	−0.01	0.04	0.24	−0.26	0.8	7.6	<0.01

Table 7, part 1: Performance persistence with three-year ranking periods.

Table 7, part 2: Performance persistence with one-year ranking periods.

7 Short summary

- The paper studies whether mutual fund stars have genuine skill or merely extreme luck.
- The data cover U.S. open-end domestic equity mutual funds from 1975 to 2002.
- The baseline sample includes 1,788 funds with at least 60 monthly observations.
- The key methodological contribution is a cross-sectional bootstrap of ranked alphas and ranked alpha t-statistics.
- The bootstrap imposes zero true alpha and asks how extreme the best funds should look by chance.
- The top part of the fund distribution contains managers whose net alphas are too high to be explained by luck alone.
- The evidence of skill is strongest among growth-oriented funds.
- Many below-median and left-tail funds underperform net of expenses and trading costs.
- Pre-cost stockholdings-based tests show broader stock-picking ability but also reveal that costs consume much of the value.
- The paper finds evidence of persistence among top-ranked funds, especially once the bootstrap replaces standard normal inference.
- The central lesson is that mutual fund evaluation requires a luck benchmark for the sorted cross section, not just a single-fund t-test.

8 Conclusion

8.1 Core conclusion

Kosowski, Timmermann, Wermers, and White show that a nontrivial minority of mutual fund managers possess stock-picking skill that survives expenses and trading costs. This conclusion is not based on a casual look at the best realized returns; it comes from comparing the actual right tail of the fund-performance distribution with a bootstrapped no-skill distribution.

8.2 Economic intuition

In a large fund universe, luck alone produces some spectacular-looking winners. The difficulty is to determine whether the realized right tail is more extreme than this luck benchmark. The paper's bootstrap creates that benchmark while preserving the observed nonnormality, fund risk-taking, and fund history lengths. When the observed top funds exceed the bootstrap-implied lucky outcomes, this is evidence of skill.

8.3 Incremental contribution revisited

The paper's contribution is methodological and substantive. Methodologically, it shows how to evaluate mutual fund stars without assuming a simple normal distribution for ranked alphas.

Substantively, it shows that the average-fund result misses important heterogeneity: most funds do not add value, but some right-tail growth funds do.

8.4 Implications

The findings matter for investors, researchers, and the mutual fund industry. Investors should not infer skill from a high realized alpha without accounting for the fact that many funds were examined. Researchers should use cross-sectional inference when evaluating ex post performance rankings. The industry result is nuanced: active management often fails to cover costs, but it is not universally valueless.

8.5 Limitations

The bootstrap separates skill from luck statistically, but it does not tell investors perfectly which fund will be skilled before the fact. It also depends on the maintained performance model, the fund sample, and the assumption that resampled residuals capture the relevant no-skill distribution. Loads, taxes, manager turnover, and changing fund strategies also complicate the practical investor interpretation.

8.6 Takeaway

The enduring takeaway is that mutual fund stars can exist, but proving that they exist requires a careful no-skill benchmark. After building that benchmark with a bootstrap, the paper finds that some high-performing funds really do pick stocks well enough to cover their costs, while many other funds destroy value through expenses and poor net performance.